

The Generation-5 Tumour-Organoid Model

Theory, Governing Equations, Lineage from G4, and Ablation Logic

Motor-clutch collagen-remodelling project

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Abstract

Generation 5 (G5) scales the validated single-cell motor-clutch and bead-spring collagen framework to a multi-cellular tumour organoid. The clean conceptual change is one cell to many mechanically interacting cell disks. This requires organoid geometry, cell-cell repulsion and adhesion, repeated cell-ECM contact bookkeeping, and collective migration metrics. Tensile strain stiffening, leader-cell heterogeneity, and crosslink topology plasticity are separate hypotheses that should be activated by ablation rather than hidden in the baseline. The model is two-dimensional and coarse-grained; its parameters are mechanism-first until calibrated against organoid experiments.

KEY EQUATION

$$\zeta_b \dot{\mathbf{r}}_i = \mathbf{F}_i^{\text{stretch}} + \mathbf{F}_i^{\text{bend}} + \mathbf{F}_i^{\text{xlink}} + \sum_a \mathbf{F}_{ia}^{\text{rep}} + \sum_a \mathbf{F}_{ia}^{\text{cell} \rightarrow \text{ECM}}. \quad (1)$$

The collagen equation remains the common backbone. G5 changes the number of cells and the coupling terms, not the unit system.

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1 Lineage: what is inherited and what is new

G5 inherits the G2/G4 bead-chain geometry, axial stretch, discrete bending, compression softening, outer-boundary anchoring, permanent elastic crosslinks, local material-point contact, and optional Bell motor-clutch kinetics. The genuinely required additions are multiple rigid cell disks, cell-cell mechanics, per-cell contact ownership, and collective outcomes.

Stage	New mechanism	Primary question
G5A	N cell disks plus cohesion	Is the organoid stable before cell-ECM pulling?
G5B	Many cells pull one shared ECM	Does collective contraction increase radial collagen alignment?
G5C	Tensile strain stiffening	Does nonlinear recruitment extend the deformation halo?

G5D	Reaction-driven cell motion	Does cell–cell adhesion select collective versus single-cell escape?
G5E	Crosslink rupture and re-welding	Does topology change preserve remodelling after unloading?

The clean G5 baseline is multicellular disks, cell–cell adhesion, inherited elastic collagen, and inherited clutch mechanics. Strain stiffening, leader cells, and crosslink plasticity are explicit switches. This separation keeps G4 collagen calibration conceptually distinct from the multicellular consequences introduced in G5.

2 Inherited collagen mechanics

Every collagen fibre is a connected sequence of beads. Visible lines join consecutive beads, and crosslinks join material points on different fibres. The outer boundary is anchored so local remodelling is not confused with rigid translation of the whole network.

For non-fixed beads,

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \frac{\Delta t}{\zeta_b} \mathbf{F}_i^{\text{total}}. \quad (2)$$

There is no inertia and no standard-linear-solid element. Drag controls transient speed; elastic terms store energy.

For an axial bond,

$$e_{ij} = \ell_{ij} - \ell_{ij}^0, \quad \mathbf{F}_{i \leftarrow j}^{\text{stretch}} = k_{ij} e_{ij} \hat{\mathbf{e}}_{ij}, \quad (3)$$

$$k_{ij} = \begin{cases} EA/\ell_{ij}^0, & e_{ij} \geq 0, \\ \rho EA/\ell_{ij}^0, & e_{ij} < 0, \end{cases} \quad \hat{\mathbf{e}}_{ij} = (\mathbf{r}_j - \mathbf{r}_i)/\ell_{ij}. \quad (4)$$

Compression remains soft ($\rho < 1$), representing microbuckling rather than an equally stiff compression spring.

For bending,

$$\Delta \kappa_i = (\mathbf{r}_{i-1} - 2\mathbf{r}_i + \mathbf{r}_{i+1}) - \kappa_i^0, \quad (\mathbf{F}_{i-1}^b, \mathbf{F}_i^b, \mathbf{F}_{i+1}^b) = k_b(-\Delta \kappa_i, 2\Delta \kappa_i, -\Delta \kappa_i). \quad (5)$$

A freely hinged crosslink joining two material points obeys

$$\mathbf{p}_a = (1 - \alpha_a)\mathbf{r}_i + \alpha_a\mathbf{r}_j, \quad (6)$$

$$\mathbf{F}_x = k_x[(\mathbf{p}_b - \mathbf{p}_a) - \mathbf{d}_{x,0}]. \quad (7)$$

The link transmits separation force without prescribing the crossing angle.

3 G5A: multicellular scaffold and resting balance

Cell centres are placed on a hexagonal lattice with rest pitch $d_0 = 18 \mu\text{m}$. Each rigid disk has radius $R_c = 9 \mu\text{m}$, and centres inside the organoid target radius are retained. Collagen is excluded from the union of cell disks plus a clearance region.

Let $d = \|\mathbf{c}_b - \mathbf{c}_a\|$ and $\hat{\mathbf{e}}_{ab} = (\mathbf{c}_b - \mathbf{c}_a)/d$. The pairwise force on cell a from cell b is

KEY EQUATION

$$\mathbf{F}_{a \leftarrow b}^{cc} = \begin{cases} -k_{\text{rep}}(d_0 - d)\hat{\mathbf{e}}_{ab}, & d < d_0, \\ k_{\text{adh}}s_{ab}(d - d_0)\hat{\mathbf{e}}_{ab}, & d_0 \leq d < d_0 + r_{\text{adh}}, \\ \mathbf{0}, & d \geq d_0 + r_{\text{adh}}, \end{cases} \quad s_{ab} = \min(s_a, s_b). \quad (8)$$

Repulsion prevents overlap and finite-range adhesion supplies coarse tissue cohesion. The strength k_{adh} is an effective organoid-scale parameter, not a direct cadherin single-bond stiffness. At $d = d_0$, the pair force vanishes.

Cell–collagen exclusion generalizes the G4 penalty to all cells:

$$\delta_{ia} = \max(0, R_c + c - \|\mathbf{r}_i - \mathbf{c}_a\|), \quad \mathbf{F}_i^{\text{rep}} = \sum_a k_{\text{rep,ECM}} \delta_{ia} \frac{\mathbf{r}_i - \mathbf{c}_a}{\|\mathbf{r}_i - \mathbf{c}_a\|}. \quad (9)$$

The G5A gate is a stable non-overlapping organoid that neither collapses nor disperses before active traction is introduced.

4 G5B: fixed organoid pulling a shared collagen network

For cell a and fibre material point $\mathbf{p}_{af}$, contact is eligible only inside a finite surface shell. Gaussian weights are normalized over the eligible set $C(a)$:

KEY EQUATION

$$s_{af} = \|\mathbf{p}_{af} - \mathbf{c}_a\| - R_c, \quad 0 \leq s_{af} \leq w_c, \quad (10)$$

$$w_{af} = \frac{\exp(-s_{af}^2/\sigma_c^2)}{\sum_{g \in C(a)} \exp(-s_{ag}^2/\sigma_c^2)}. \quad (11)$$

The inward pull from cell a is

$$\mathbf{F}_{af}^{\text{cell} \rightarrow \text{ECM}} = F_a w_{af} \frac{\mathbf{c}_a - \mathbf{p}_{af}}{\|\mathbf{c}_a - \mathbf{p}_{af}\|}. \quad (12)$$

Linear shape functions distribute this point force to the two host beads. A non-contact fibre receives no second Gaussian pull; it can move only through crosslink force transmission and its own stretching and bending. Removing crosslinks is therefore the causal negative control.

Radial alignment is measured on each fibre edge by

KEY EQUATION

$$S_{r,e} = 2(\mathbf{t}_e \cdot \hat{\mathbf{e}}_{r,e})^2 - 1, \quad S_r(q) = \frac{1}{N_q} \sum_{e \in q} S_{r,e}. \quad (13)$$

$S_r = +1$ is radial, $S_r = 0$ isotropic, and $S_r = -1$ tangential. Reporting distance shells separates near-field reorientation from a whole-network average.

5 G5C: tensile strain-stiffening ablation

The optional nonlinear constitutive law changes only tensile bonds:

KEY EQUATION

$$\varepsilon = \frac{e}{\ell_0}, \quad k_t(\varepsilon) = k_{t,0} \min\left(C_{\max}, \exp\frac{\varepsilon}{\varepsilon_{\text{ref}}}\right) \quad (\varepsilon \geq 0), \quad k(\varepsilon < 0) = \rho k_{t,0}. \quad (14)$$

Current defaults use $\varepsilon_{\text{ref}} = 0.06$ and $C_{\max} = 12$ as exploratory guards. A taut bond becomes stiffer and can recruit a longer load-bearing tract. The clean comparison switches strain stiffening off and on while holding network, pull, cells, and seed fixed. If most strains remain only one to five percent, the nonlinear branch is only weakly engaged; a small effect is then a statement about the current loading regime, not a rejection of collagen nonlinearity.

6 Motor-clutch kinetics in a multicellular setting

When clutch dynamics are enabled, every cell is divided into angular contact sectors and every occupied sector contains an effective clutch bundle. G5 reuses G4 kinetics but repeats them over many cells and preserves site ownership for reaction-force bookkeeping.

Motor loading follows

KEY EQUATION

$$v_{\text{act}} = v_0 \max\left(0, 1 - \frac{F_{\text{site}}}{F_{\text{stall}}}\right), \quad \dot{x} = \max(0, v_{\text{act}} - v_{\text{sub}}), \quad f_j = k_c x_j. \quad (15)$$

Bell slip-bond rupture and rebinding are

$$k_{\text{off}}(f) = k_{\text{off}}^0 \exp\left(\frac{f}{F_b}\right), \quad P_{\text{off}} = 1 - \exp[-k_{\text{off}}(f)\Delta t], \quad P_{\text{on}} = 1 - \exp(-k_{\text{on}}\Delta t). \quad (16)$$

The independent hypothesis uses

$$F_{\text{site}} = \sum_{j \in \text{bound}} k_c x_j, \quad (17)$$

whereas the equal-load-sharing hypothesis uses

$$f_i = \frac{F_{\text{site}}}{i}, \quad r_i = i k_{\text{off}}^0 \exp\left(\frac{F_{\text{site}}}{i F_b}\right). \quad (18)$$

An empty angular sector cannot bind or load a clutch. This active-mask guard prevents phantom traction against non-existent collagen. Bell rupture of a cell–fibre clutch produces adhesion slippage; Bell rupture of a fibre–fibre crosslink in G5E changes ECM topology. They are separate processes with separate parameters.

7 G5D: collective invasion versus single-cell escape

Every rigid cell disk moves under its cell–ECM reaction force and cell–cell mechanics:

KEY EQUATION

$$\zeta_c \dot{\mathbf{c}}_a = \mathbf{F}_a^{\text{reaction}} + \sum_{b \neq a} \mathbf{F}_{a \leftarrow b}^{cc}, \quad \mathbf{F}_a^{\text{reaction}} = - \sum_{s \in a} \mathbf{F}_s^{\text{cell} \rightarrow \text{ECM}}. \quad (19)$$

A cell reels collagen inward; the opposite reaction tends to pull the cell toward the gripped ECM. Adhesion resists separation. The velocity is capped when necessary:

$$\mathbf{v}_a = \mathbf{F}_a^{\text{net}} / \zeta_c, \quad \|\mathbf{v}_a\| > v_{\text{max}} \Rightarrow \mathbf{v}_a \leftarrow v_{\text{max}} \frac{\mathbf{v}_a}{\|\mathbf{v}_a\|}. \quad (20)$$

High adhesion favours cohesion; low adhesion permits cell–ECM reaction to separate individual cells more easily. “Invasion” in this model means outward displacement arising from this force competition, not a complete biological tissue-invasion programme.

Optional leader heterogeneity is imposed on selected outer cells:

$$s_a = s_{\text{leader}} < 1, \quad k_{\text{adh}, ab} = k_{\text{adh}} \min(s_a, s_b), \quad F_a^{\text{pull}} = \lambda_a F^{\text{pull}}. \quad (21)$$

This is a modelling hypothesis; the current outermost-fraction selection is deterministic and is not an emergent EMT mechanism.

8 G5E: crosslink topology plasticity

Loaded fibre–fibre crosslinks may rupture with a Bell-type hazard:

KEY EQUATION

$$k_{\text{off}, x}(F_x) = k_{\text{off}, x}^0 \exp\left(\frac{F_x}{F_{b, x}}\right), \quad P_{\text{rupture}} = 1 - \exp[-k_{\text{off}, x}(F_x) \Delta t_x]. \quad (22)$$

Current defaults $k_{\text{off}, x}^0 = 2 \times 10^{-4} \text{ s}^{-1}$ and $F_{b, x} = 3 \text{ nN}$ are explicit assumptions pending calibration. When a new crossing is accepted, its current configuration becomes stress free:

$$\mathbf{d}_{x,0} \leftarrow \mathbf{p}_b(t) - \mathbf{p}_a(t), \quad \text{with re-weld acceptance probability } p_{\text{reform}}. \quad (23)$$

Changing which fibres are welded and where the new rest state is created is topology-changing remodelling, not SLS stress relaxation.

Topology memory and residual response are summarized by

$$\kappa_{\text{topo}} = 1 - \frac{|L_0 \cap L_T|}{|L_0|}, \quad (24)$$

$$\kappa_{\text{rms}} = u_{\text{residual}} / u_{\text{peak}}, \quad (25)$$

$$\kappa_{\text{order}} = \frac{S_{\text{final}} - S_0}{S_{\text{peak}} - S_0}. \quad (26)$$

κ_{topo} is the cleanest topology metric. Residual displacement alone can be confounded by slow elastic relaxation and near-zero modes.

9 Ablation logic

An ablation is a matched comparison in which one mechanism is removed or activated while the initial network, seed, duration, and other parameters remain fixed:

KEY EQUATION

$$\Delta Y_{(-m)} = Y_{\text{baseline}} - Y_{\text{mechanism } m \text{ removed}}. \quad (27)$$

Y may be invasion distance, cohesion, radial order, displacement reach, traction, or topology memory. The comparison attributes effects inside the model; it does not by itself establish an experimental effect size.

Ablation	Mechanism isolated	Primary readout
No clutch	Active cell–fibre loading	Invasion, traction, and slip events.
No adhesion	Organoid cohesion	Dispersion, escaped-cell fraction, and collective integrity.
No crosslinks	Network force transmission	Indirect displacement, radial order, and invasion.
Stiffening off/on	Nonlinear tensile recruitment	Alignment and displacement reach.
Plasticity off/on	Topology memory	κ_{topo} and post-unload persistence.
Leaders off/on	Prescribed cell heterogeneity	Leader displacement and follower cohesion.

In the current matched two-hour runs, approximate reported displacement/invasion is $25 \mu\text{m}$ for the clutch baseline, $4.7 \mu\text{m}$ with no clutch, $68 \mu\text{m}$ with no adhesion but worse cohesion, $7.7 \mu\text{m}$ with no crosslinks, $24 \mu\text{m}$ with strain stiffening, and $25 \mu\text{m}$ with plasticity. These are single-model outputs, not experimental effect sizes; ensembles and calibration are required.

10 Parameter map and evidence boundaries

Parameter	Current default	Role and status
Cell radius R_c	$9 \mu\text{m}$	Single-cell scale, not a fit to one cell line.
Organoid centre radius	$63 \mu\text{m}$	Two-dimensional core geometry.
Cell spacing d_0	$18 \mu\text{m}$	Touching hexagonal disks and zero pair force.
Cell repulsion / adhesion	$40 / 6 \text{ nN}/\mu\text{m}$	Effective non-overlap and cohesion parameters.
Adhesion range	$6 \mu\text{m}$	Finite-neighbour interaction assumption.
Collagen modulus	3 MPa	Inherited softened mechanism-first choice.
Crosslink stiffness	$10 \text{ nN}/\mu\text{m}$	Effective link penalty.
Fibres / domain	$420 / 440 \mu\text{m}$	Organoid-scale computational network.
Crosslink retain fraction	0.85	Generator connectivity target, not G4's probability mechanism.
Pull per cell	12 nN	Constant-pull branch only.
Bead time step	0.05 s	Inherited numerical resolution.
Cell drag / speed cap	$400 \text{ nN s}/\mu\text{m} / 0.02 \mu\text{m/s}$	Effective mobility and numerical/biological guard.

The inherited effective clutch defaults are $N = 12$, $k_c = 2 \text{ nN}/\mu\text{m}$, $k_{\text{on}} = 0.055 \text{ s}^{-1}$, $k_{\text{off}}^0 = 0.018 \text{ s}^{-1}$, $F_b = 1.5 \text{ nN}$, $v_0 = 0.025 \mu\text{m/s}$, and $F_{\text{stall}} = 8 \text{ nN/site}$. G4 crosslink probability and G5 crosslink retention fraction are different constructions and should not be compared as one biological density.

G5 remains a 2-D rigid-disk model without cell deformation, a nucleus, division, death, growth pressure, proteolysis, nutrient fields, three-dimensional pore confinement, or an emergent organoid polarity field. Several coefficients are exploratory. Matched ensembles and sensitivity analysis are required before biological prediction.

Key references

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Claim boundary. G5 is a controlled 2-D mechanism platform showing how inherited clutch traction, cell–cell cohesion, and crosslinked collagen jointly produce different organoid behaviours. It is not yet a predictive digital twin of tumour invasion.